

ENTERPRISE ARCHITECTURE (EA) AND DATABASE PROGRAMMING

LECTURE 4

Azadeh Sarkheyli
(azadeh.sarkheyli@hh.se)

CONTENTS:

- Architecture Methods and Frameworks
- Description Languages

ARCHITECTURE METHODS AND FRAMEWORKS

- The Zachman Framework
- The Open Group Architecture Framework (TOGAF)

ARCHITECTURE METHODS AND FRAMEWORKS

The Zachman Framework

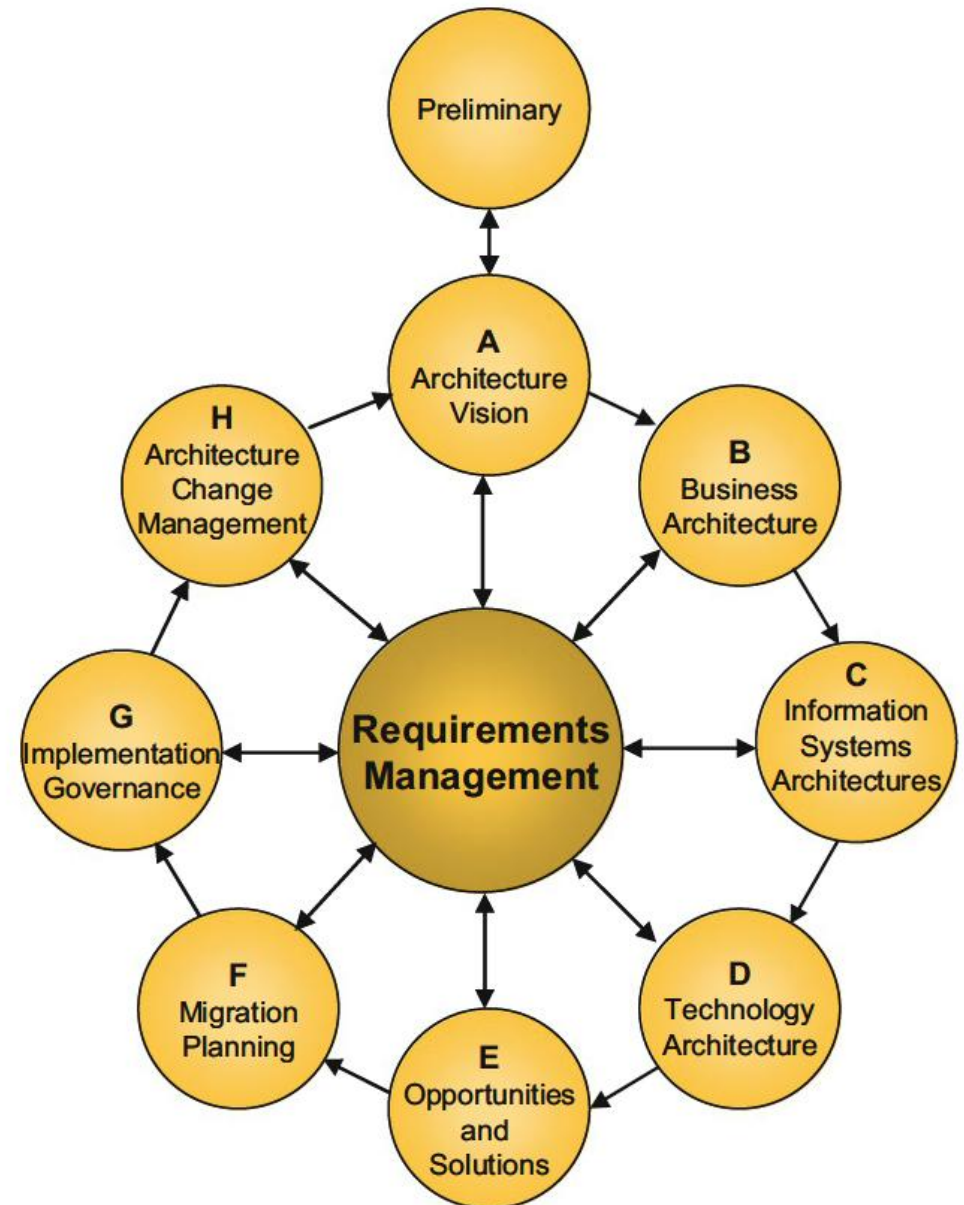
	What?	How?	Where?	Who?	When?	Why?	
Planner							Scope (contextual)
Owner							Enterprise model (conceptual)
Designer							System model (logical)
Builder							Technology model (physical)
Sub- contractor							Detailed representations (out of context)
	Data	Function	Network	People	Time	Motivation	

Fig. 2.6 The Zachman framework (Zachman 1987)

ARCHITECTURE METHODS AND FRAMEWORKS

The Open Group Architecture Framework

Fig. 2.8 TOGAF
architecture development
method (The Open Group
2011)



ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

Discussion

According to the below papers (links) describe **what are the differences between TOGAF and Zachman?**

- ❖ <https://www.bcs.org/articles-opinion-and-research/a-comparison-of-the-top-four-enterprise-architecture-frameworks/>
- ❖ <https://www.acte.in/togaf-vs-zachman-article>
- ❖ <https://www.bmc.com/blogs/togaf-vs-zachman/>



DESCRIPTION LANGUAGES

- Integrated DEFinition Methods (**IDEF**)
- The Business Process Modelling Notation (**BPMN**)
- The Unified Modeling Language (**UML**)

DESCRIPTION LANGUAGES : IDEF

- ❑ **IDEF** is the name of a family of languages used to perform enterprise modelling and analysis (see <http://www.idef.com/>)



IDEF Family of Methods

A Structured Approach to Enterprise Modeling & Analysis

IDEF.com was developed and is maintained by [Knowledge Based Systems, Inc. \(KBSI\)](#), the developers of the next generation IDEF methods: the IDEF3 Process Flow and Object State Description Capture Method, the IDEF4 Object-Oriented Design Method, and the IDEF5 Ontology Description Capture Method. IDEF, initially abbreviation of ICAM Definition, renamed in 1999 as Integration DEFinition, refers to a family of modeling languages in the field of systems and software engineering. They cover a wide range of uses, from functional modeling to data, simulation, object-oriented analysis/design and knowledge acquisition. These definition languages were developed under funding from U.S. Air Force and are heavily used by them, as well as other military and United States Department of Defense (DoD) agencies, are also used by many corporation for business process capture and improvement.

DESCRIPTION LANGUAGES : IDEF

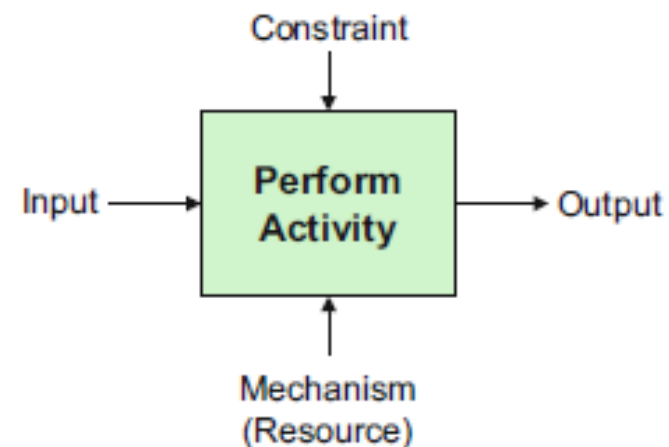
IDEF0, IDEF3, and IDEF1X ('the core') are the most commonly used. Their scope covers:

- ❖ **Functional modelling, IDEF0:** The idea behind IDEF0 is to model the elements controlling the execution of a function, the actors performing the function, the objects or data consumed and produced by the function, and the relationships between business functions (shared resources and dependencies).
- ❖ **Process modelling, IDEF3:** IDEF3 captures the workflow of a business process via process flow diagrams. These show the task sequence for processes performed by the organisation, the decision logic, describe different scenarios for performing the same business functions, and enable the analysis and improvement of the workflow.
- ❖ **Data modelling, IDEF1X:** IDEF1X is used to create logical data models and physical data models by the means of logical model diagrams, multiple IDEF1X logical subject area diagrams, and multiple physical diagrams.

DESCRIPTION LANGUAGES : IDEF

- There are five elements to the IDEF0 functional model (see Fig. 2.10): the activity (or process) is represented by boxes, inputs, outputs, constraints, or controls on the activities, and mechanisms that carry out the activity.

Fig. 2.10 IDEF0 representation



DESCRIPTION LANGUAGES : IDEF

The **IDEF3** Process Description Capture Method provides a mechanism for collecting and documenting processes. There are two IDEF3 description modes:

- Process flow diagrams
- Object state transition network diagrams

The IDEF3 term for elements represented by boxes is a **Unit Of Behaviour (UOB)**. The arrows (links) tie the boxes (activities) together and define the logical flows. The smaller boxes define junctions that provide a mechanism for introducing logic to the flows.

DESCRIPTION LANGUAGES : IDEF

❑ Process flow diagrams:

A process flow description captures knowledge of 'how things work' in an organization, e.g., the description of what happens to a part as it flows through a sequence of manufacturing processes.

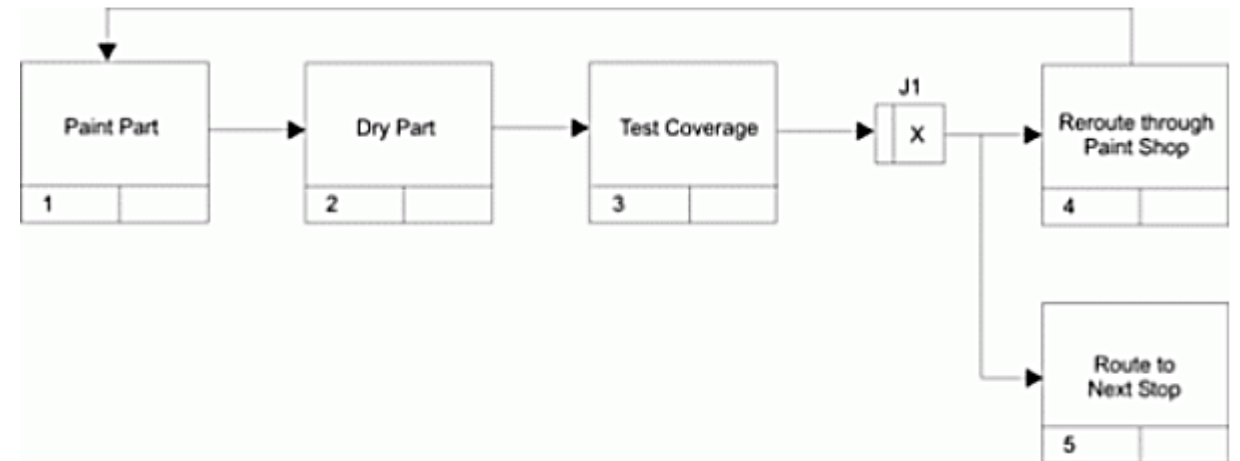


Figure 1: IDEF3 Process Description Diagram

DESCRIPTION LANGUAGES : IDEF

❑ Object state transition network diagrams

The object state transition network description summarizes the allowable transitions an object may undergo throughout a particular process.

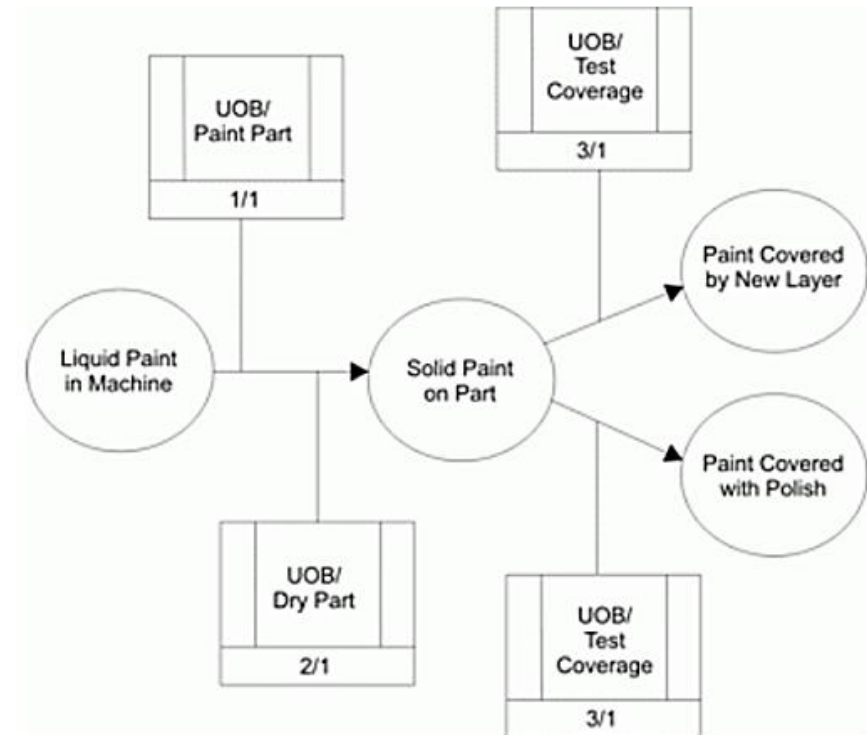


Figure 2: IDEF3 Object State Transition Network Diagram

DESCRIPTION LANGUAGES : IDEF

- The **IDEF family** provides support for the modelling of several architectural views. However, there are no communication mechanisms between models. The fact that they are isolated hinders the visualisation of all models as interrelated elements of an architectural system. This also means that a switch between views is not possible.
- **IDEF** is widely used in the industry. This indicates that it satisfies the needs of the users within acceptable limits.
- The **IDEF family** is subject to a continuous process of development and improvement. Still, IDEF0, IDEF1X, and IDEF3 are rather stable and rigid languages, and IDEF0 and IDEF1X have been published as standards of the National Institute of Standards and Technology.

DESCRIPTION LANGUAGES : BPMN

- The Business Process Modelling Notation (BPMN) was developed by the Business Process Management Initiative (BPMI), which has since merged with the Object Management Group.
- The BPMN standard (Object Management Group 2013) specifies a graphical notation that serves as a common basis for a variety of business process modelling and execution languages.
- As the name already indicates, BPMN is restricted to process modelling; applications or infrastructure are not covered by the language.
- The main purpose of BPMN is to provide a uniform notation for modelling business processes in terms of activities and their relationships (Fig. 2.11).

DESCRIPTION LANGUAGES : BPMN

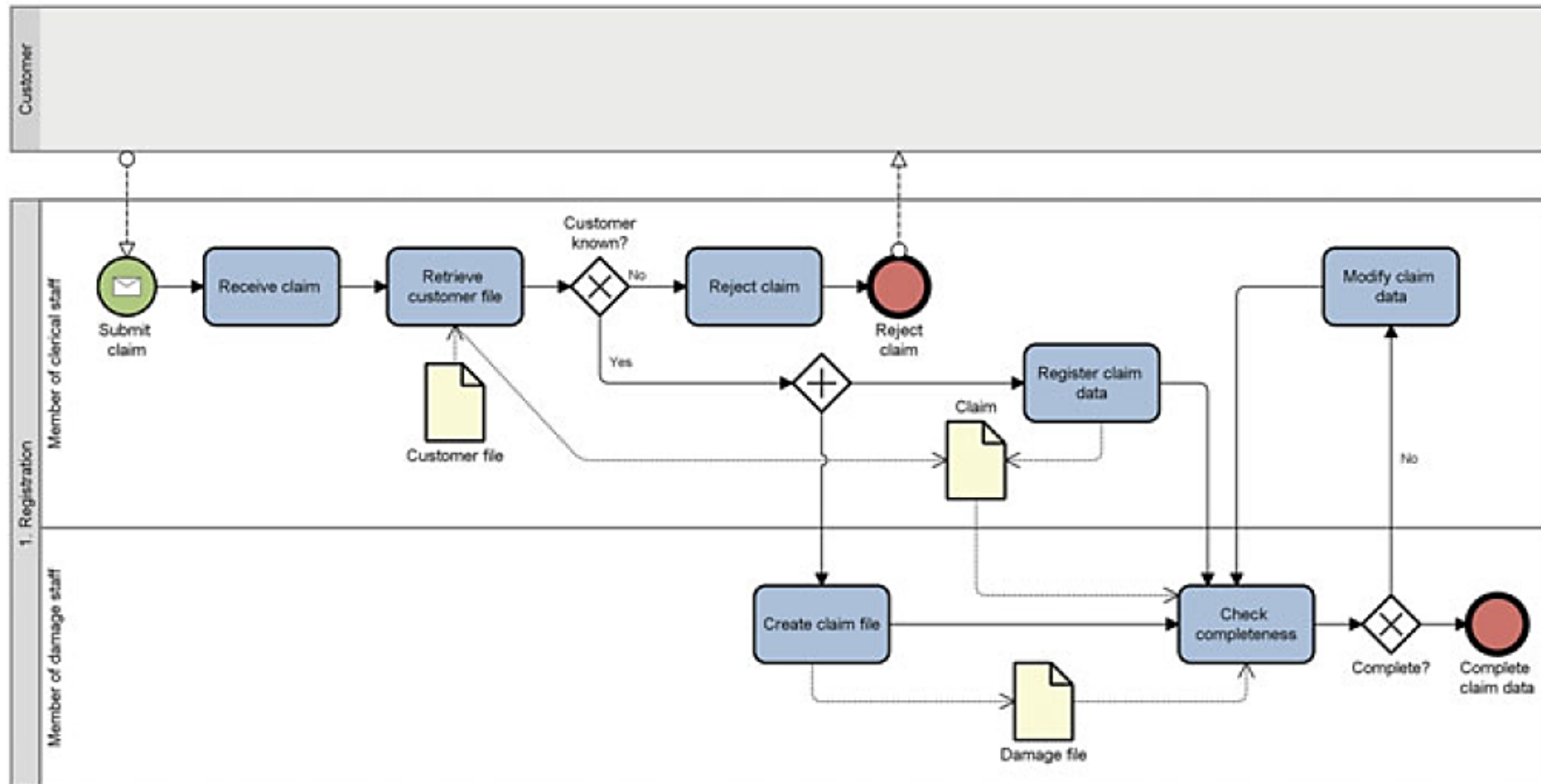


Fig. 2.11 Example model in BPMN

DESCRIPTION LANGUAGES : UML

- The Unified Modeling Language (UML) is currently the most important industry-standard language for specifying, visualising, constructing and documenting the artefacts of software systems.
- The language's development is managed by the Object Management Group (OMG).
- UML is intended to be used by system designers. Consequently, UML models are only clear to those who have a sound background in computer science, in particular in object orientation.
- UML models should be sufficiently understandable for illustrative and explanatory purposes to business engineers and organization specialists. Although UML was originally developed for the design of object-oriented software, its use has expanded to other areas, including architecture modelling.

DESCRIPTION LANGUAGES : UML

- **UML** is a disturbingly rich combination of 13 sublanguages each having its own (sub)scope of the complete UML scope, and each with its own diagram to model a specific aspect of a (software) system. The 13 diagrams can be grouped in three categories:
 - **Structure:** package diagrams, class diagrams, object diagrams, composite structure diagrams;
 - **Behaviour:** use case diagrams, state diagrams, sequence diagrams, timing diagrams, communication diagrams, activity diagrams, interaction overview diagrams;
 - **Implementation:** component diagrams, deployment diagrams.

DESCRIPTION LANGUAGES

Suitability for Enterprise Architecture

In the previous sections, we have given an overview of several languages for modelling in the area of **organizations, business processes, applications and technology**. It is clear that none of these has succeeded in becoming 'the language' that can cover all domains. In general, there are a number of aspects on which almost all of these languages score low:

- ❑ The relations between domains (views) is poorly defined, and the models created in different views are not further integrated.
- ❑ Most languages have a weak formal basis and lack a clearly defined semantics.
- ❑ Most languages miss the overall architectural vision and are confined to either the business or the application and technology subdomains.

ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

Discussion

According to the below paper describe **what are the differences between BPMN, UML, and IDEF?**

- ❖ Entringer, T. C., Ferreira, A. D. S., & Nascimento, D. C. D. O. (2021). Comparative analysis of the main business process modeling methods: a bibliometric study. *Gestão & Produção*, 28.



REFERENCES

- Lankhorst, M. (2017). Enterprise architecture at work (ISBN 978-3-662-53933-0). Berlin: Springer.
- Entringer, T. C., Ferreira, A. D. S., & Nascimento, D. C. D. O. (2021). Comparative analysis of the main business process modeling methods: a bibliometric study. *Gestão & Produção*, 28.

